

WEEK 2 REPORT

Project Title: Diabetes / Heart Disease Risk Predictor

Intern Name: Yuvashree D

Domain: Machine Learning

Week 2 Objective

To preprocess the datasets, train machine learning models using cleaned data, and evaluate their performance using basic classification metrics.

Tasks Completed

1. Feature Engineering

- Loaded Diabetes and Heart Disease datasets.
- Checked for missing values.
- Removed duplicate records.
- Applied StandardScaler to scale numerical features.
- Saved diabetes_cleaned.csv and heart_cleaned.csv.

2. Model Training

Trained Logistic Regression and Random Forest models and saved them as .pkl files.

3. Performance Evaluation

Evaluated the models using Accuracy Score, Confusion Matrix, and Classification Report.

Tools Used

Python, Pandas, NumPy, Scikit-learn, Joblib

Files Created

feature_engineering.py
model_training.py
performance_metrics.py
diabetes_cleaned.csv
heart_cleaned.csv
Model (.pkl) files

Conclusion

The datasets were successfully preprocessed, the models were trained, and their performance was evaluated. This completed the Week 2 objectives of the internship project.